Md Sadekul Islam

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RESEARCH INTERESTS

- **Engineering Design & Product Development:** CAD/CAE, Finite Element Analysis (FEA), Design for Manufacturing (DFM), mechanism design, and material selection for complex consumer products.
- **Advanced Manufacturing:** additive manufacturing (3D printing), and Industry 4.0 process optimization.
- **Simulation & Optimization:** Deep Reinforcement Learning (DRL), Graph Neural Networks (GNNs), and genetic algorithms for facility layout; discrete-event simulation (DES) and digital twin modeling.
- **Biomechanics & Human-Centered Design:** Assistive and wearable devices, ergonomics, safety management, and human-machine interaction.

EDUCATION

Bangladesh University of Engineering and Technology (BUET) Sept 2026

M.Engg. in Advanced Engineering Management, Dhaka, Bangladesh | GPA 3.45/4.00

Thesis: Integrating Evolutionary Algorithm and Deep Reinforcement Learning for Efficient Facility Layout Design.
Supervisor: Dr. Ridwan Al Aziz, Associate Professor, Dept. of Industrial and Production Engineering.

- Developed a two-stage pipeline in which a genetic algorithm generates feasible layouts that a Rainbow DQN agent refines, using a 12-component composite reward, action masking, dimension-aware multi-cell overlap prevention, and a GNN encoder pre-trained on synthetic flow graphs.
- Evaluated on a 26-machine industrial layout from a Bangladeshi manufacturer, with sensitivity analyses over reward weights (21-run design at two search budgets), grid resolution (9, 11, 13) and flow density (six flow configurations).
- Documented the open problems (training instability, residual overlaps, unreliable outcomes under very sparse flow data) and the next steps that frame my planned doctoral work: multi-seed evaluation, longer training, and constraint-aware repair.

Khulna University of Engineering and Technology (KUET) March 2019

B.Sc. in Industrial and Production Engineering, Khulna, Bangladesh | GPA 3.59/4.00 | Ranked 11th of 54

Thesis: Improvement of Production Line Efficiency by Using Tecnomatix Simulation Software and RPWM Line Balancing Technique. **Supervisor:** Dr. Mahmud Parvez.

PUBLICATIONS

- Islam, M. S., Sarker, S., & Parvez, M. (2019). Production efficiency improvement by using Tecnomatix simulation software and RPWM line balancing technique: A case study. *American Journal of Industrial and Business Management, 9*(4), 809–820.
- **Manuscript in preparation:** Integrating Evolutionary Algorithm and Deep Reinforcement Learning for Efficient Facility Layout Design (based on M.Engg. thesis).

PROFESSIONAL EXPERIENCE

Walton Hi-Tech Industries PLC, Gazipur, Bangladesh Mar 2019 – Present

Mechanical Product Designer, Refrigerator R&D

- **Flagship launch:** led the design and launch (July 2023) of Walton's largest side-by-side refrigerator, the 646 L non-frost model with more than 250 parts and components.
- **Subsystem design:** designed the cabinet liner and the water-dispenser and ice-tray mechanisms in SolidWorks (tight tolerances; combined electrical, plastic, potable-water and spring components) and verified stability, reliability and performance through computer simulation and laboratory tests.

- **Design for manufacturing:** performed draft analysis (core/cavity, sliders, lifters), ran injection-mold trials on a duct cover (mold temperature, injection parameters, gate count), and verified parts with 3D-printed prototypes before mold design.
- **Benchmarking and release:** benchmarked competing designs (e.g., door-gasket profiles) and ran volume, weight and cost analyses before component design; produced 2D drawings with tolerances and materials; reviewed pilot-production samples for misalignment, gaps, leakage and color discrepancies.
- **Other products:** contributed to a combi refrigerator and a beverage cooler, validating beverage-cooler components with push-pull, drop, snap-fit and steady-state thermal simulations.
- **Analytics and decision support:** built a Power BI project-management dashboard for the Refrigerator R&D department; analyzed market trends, break-even and performance indices with the Deputy Chief Business Officer; led a team developing dashboards and KPI systems for project and employee performance tracking.

Brothers Furniture Ltd., Savar, Bangladesh March 2018

Industrial Training

Contemporary furniture manufacturing: 3D modeling and design changes in AutoCAD and SolidWorks, 3D scanning for reverse engineering, process monitoring for quality control, and exposure to CNC and robotic production.

SELECTED DESIGN PROJECTS (SEE PORTFOLIO)

- **Virtual factory design** (SketchUp, V-Ray): 3D layout models of a planned production facility with equipment, storage and safety zones to support layout, safety and human–machine interaction review.
- **Prosthetic leg** (Fusion 360, KeyShot): CAD design of a prosthetic leg assembly aimed at comfort, function and cosmetic acceptability.
- **Portable speaker** (SolidWorks, Fusion 360, KeyShot): three-component assembly without screws, designed for low weight and shock/vibration resistance; injection-molding flow analysis in Autodesk Moldflow to inform material, process and tooling choices.

HONORS AND AWARDS

- First Runner-Up, Advance Engineering Management Challenge (AEMC), KUET, February 2018
- Merit-based scholarship in each of the eight semesters of the B.Sc. program, KUET
- Souvenir for outstanding performance, Walton Hi-Tech Industries PLC

TECHNICAL SKILLS

Design and analysis: SolidWorks (3D design, motion, evaluation), ANSYS, PTC Creo, Autodesk Fusion 360, AutoCAD, KeyShot, SketchUp with V-Ray, Autodesk Moldflow

Simulation and optimization: Siemens Tecnomatix Plant Simulation, Arena, Proplanner, LINGO, TORA

Programming and ML: Python, C, C++; thesis stack: TensorFlow, PyTorch, PyTorch Geometric, Gymnasium, SimPy, NumPy, Pandas

Data and project management: Power BI, Microsoft Excel, Minitab, Oracle, Microsoft Project, Primavera P6

ENGLISH PROFICIENCY

IELTS (Nov 2023): Overall 7.0 — Listening 7.0, Reading 8.0, Writing 6.5, Speaking 6.0

References available on request.